R4R: Reproducibility for R

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Motivation

Reproducibility is a cornerstone of scientific progress, but in computational research it remains a major challenge.

Failure Rates

74% of R scripts fail to re-run (Harvard Dataverse study)

65% of R Markdown notebooks from GitHub fail to execute

Causes

- Missing or incompatible R packages
- Missing system libraries
- Removed or moved input datasets
- File/path configuration errors

Manual reconstruction of environments is time-consuming and error-prone, especially for non-technical users.

Objective

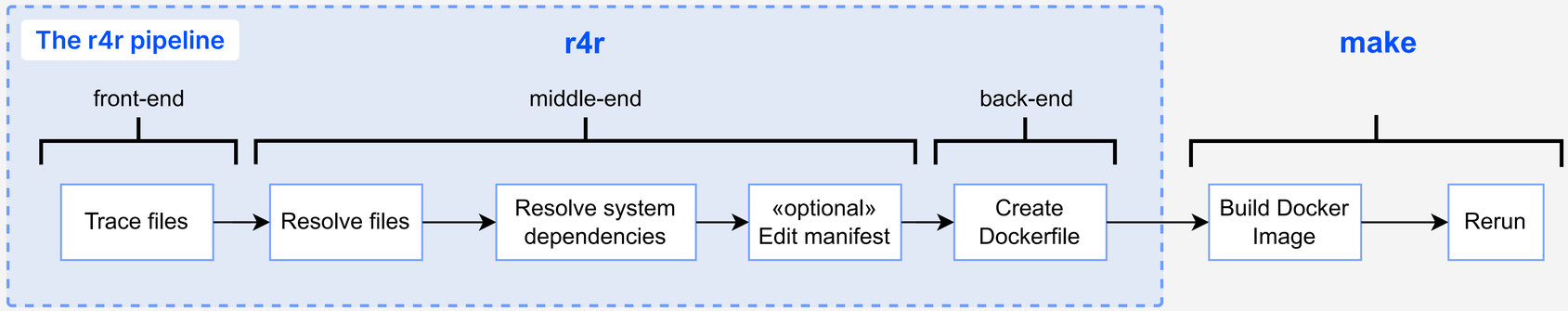
Automate the creation of **minimal, portable, inspectable, and reusable** artifacts from R notebooks, ensuring computational reproducibility.

Requirements for Reproducibility

Capture all sources of non-determinism:

- **Application-level:** file dependencies, environment variables, user/group access control
- **Language-level:** R packages (CRAN, GitHub)
- **System-level:** Debian/Ubuntu packages, system libraries

Artifacts should be: **Shareable - Inspectable - Reusable**



The R4R tool

R4R automates full environment reconstruction for R notebooks with high reproducibility rates. Produces **minimal, portable, and human-inspectable** Docker images.

Dynamic program tracing during notebook execution to detect:

- All accessed files
- System calls (e.g., `openat, execve, readlinkat, newfstatat`)
- R and system package dependencies

Design & Implementation

Currently, R4R supports:

- **Frontend:** Linux (tested on Ubuntu/Debian)
- **Middle-end:** R packages, deb packages
- **Backend:** Docker image generation

Terminal
```
$ r4r R -e 'rmarkdown::render("notebook.Rmd")'
```

R
```
r4rwrapper::r4r_traceRmd("notebook.Rmd")
```

The tool is extensible, allowing future front-end for macOS and Windows, middle-end support for other languages such as Python, and back-end for generating virtual machines.

Output:

- Docker image
- Editable manifest
- Makefile for building/running

Evaluation

Dataset: Sample of 120 open-source Kaggle notebooks

Results:

- 97.5% exact reproducibility for deterministic notebooks
- Median slowdown: 3.2× – due to tracing & package resolution
- Median build time: 32 min/notebook – mostly R package installation
- Median image size: 1.64 GB – vs 38 GB Kaggle base image

github.com/r-tooling/r4r